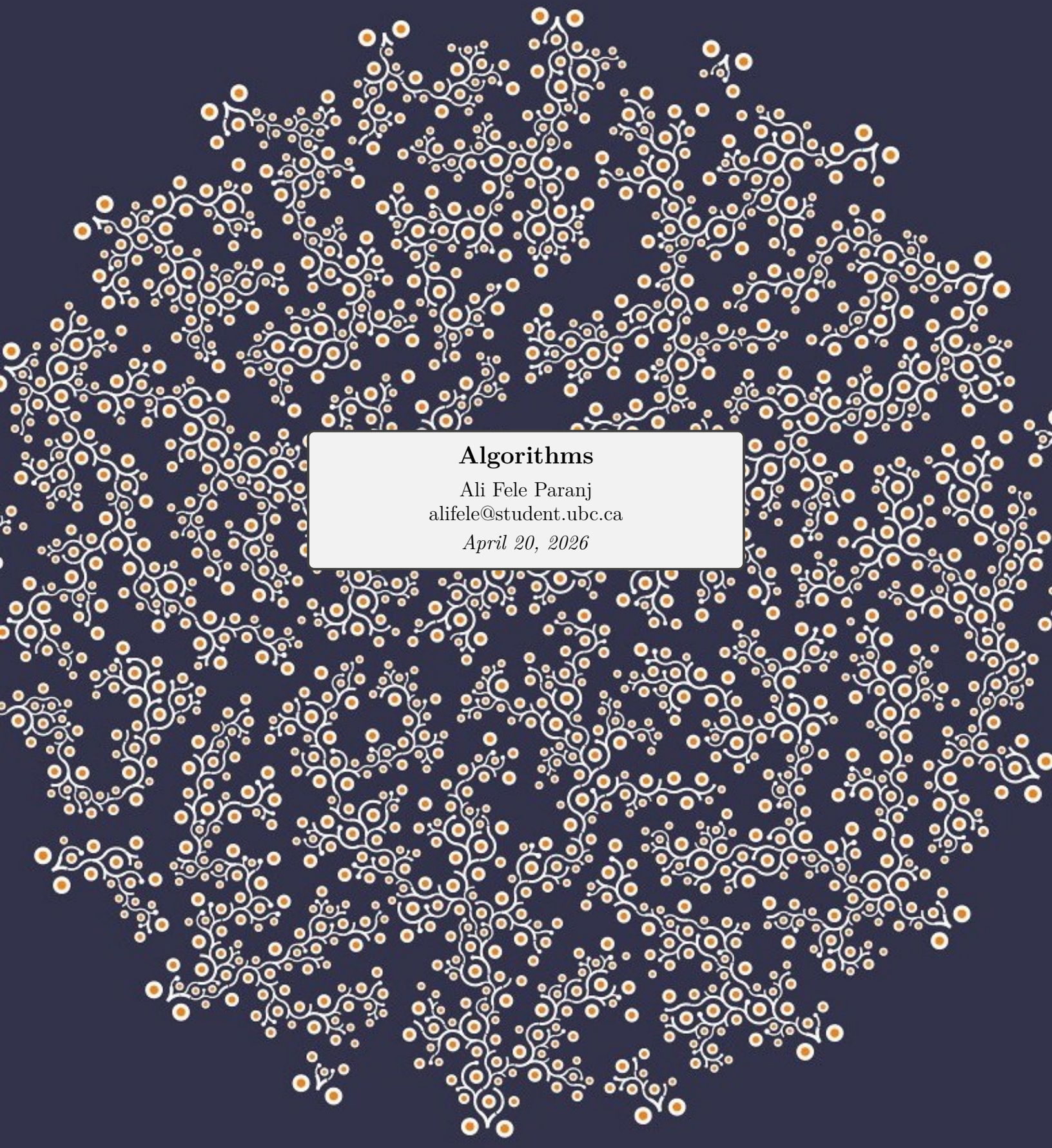

Algorithms

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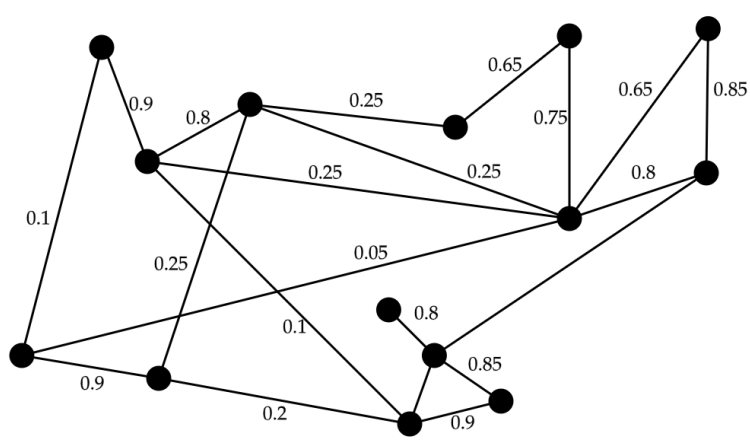
1. Classic Algorithms

1.1 Greedy Algorithm

1.1.1 Clustering Problem

Consider the following weighted graph. One can interpret the weights of the edges as the similarity between nodes. For instance, the nodes can correspond to different images in a dateset, and we draw an edge between two such images if they are comparable in some sense, where the weight of the edge is the similarity level between the images. The problem is given the weighted graph and an specified number of "*clusters*" or "*categories*", we want to group the nodes in a way that cost of the clustering is as small as possible. The *cost* of clustering is the maximum weight of edges that are between two categories.



Assume that the given weighted graph above, we want to categorize the nodes into 5 clusters. We follow the following greedy algorithm:

input: Given weighted graph G (n nodes, m edges) and number of clusters c:

do: Sort the edges e_i in decreasing order of their weight.

do: make every node to be its own cluster.

define: $C \leftarrow n$: number of current clusters

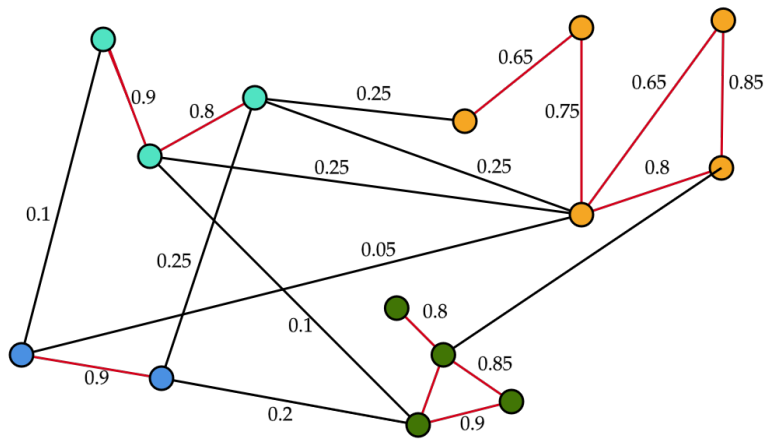
while: $C \neq c$ iterate over E

do: remove e_i from E

if: adjacent nodes p_1, p_2 to e_i are not in the same category

do: join the categories of p_1 and p_2.

do: $C \leftarrow C - 1$.



2. Nature of Computing

This chapter contains the solutions to the exercises of the book "Nature of Computing".

2.1 Chapter 1

■ **Problem 2.1 — Hand shake theorem.** Prove that in any finite graph, the number of verticies with odd degree is even.

Solution Observe that the sum of degrees of all vertices is an even number,

$$\sum_{x \in N} \deg(x) = 2|E|.$$

So there should be even number of verticies that has odd degree (other wise the sum above will be an odd number).

■ **Problem 2.2 — Pigeons and holes.** Show that in any finite simple graph with more than one vertex, there is at least one pair of vertices that have the same degree.

Solution We consider three exhaustive cases: The graph G has:

(i) no isolated nodes: Then the deg function is

$$\deg : \{n_1, \cdots, n_N\} \to \{1, \cdots, N-1\},$$

This function can not be one-to-one, because the size of the range is smaller than the size of the domain. So $\exists n_i, n_j$ such that $\deg(n_i) = \deg(n_j)$.

(ii) Exactly one isolated nodes: without loss of generality, assume n_1 is the isolated node. So deg restricted on the non-isolated nodes will be

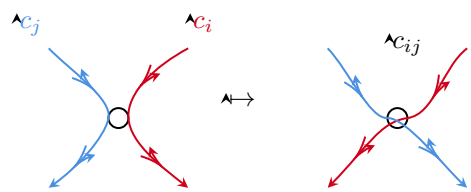
$$\deg : \{n_2, \cdots, n_N\} \to \{1, \cdots, N-2\}.$$

For the same reason as above deg is not one-to-one.

(iii) More than one isolated nodes: Then immediately we have at least two nodes that both have zero degree.

■ **Problem 2.3 — Proving Euler's Claim.** Prove that having vertices with even degree is sufficient for a connected graph to be Eulerian. *Hint: First show that if every vertex has even degree, we can cover the graph with a set of cycles such that every edge appears exactly once. Then consider combining cycles.*

Solution First, we show that if the degree of all nodes is even, then one can cover the graph with independent cycles. Starting with a non-isolated node n_1, let c_1 be a cycle that contains n_1. To show that such a cycle exists, let $\tilde{c}_1$ be a $n_1 \to \tilde{n}_1$ simple path, where $\tilde{n}_1$ is a neighbor of n_1 that is not on the path. Note that such a neighbor exists, because $\deg n_1$ is non zero and even). Let c_1 be the path that one gets by connecting the path $\tilde{c}_1$ to n_1. Now delete all the edges in c_1 from the graph G to get the new graph G_1. Observe that every node in G_1 still has even degree, with possibly more isolated nodes compared to G. We continue this process. Since G is finite, then this process will terminate. The set of paths $\{c_1, \cdots, c_n\}$ are independent simple cycles. For two cycles c_i and c_j that share a node n, we merge then according to the following diagram By



this construction c_{ij} is a bigger non-simple cycle (it is non-simple because it intersects itself). By continuing merging cycles to one another, we eventually get the Euler cycle.